

EMI OBSERVATION

x, y, z

$$x = -4$$

$$\rho = \sqrt{x^2 + y^2}$$

$$y = 11$$

$$\phi = \tan^{-1}(y/x)$$

$$z = -6$$

$$z = z$$

cylindrical

$$\begin{aligned} \rho &= \sqrt{x^2 + y^2} \Rightarrow \sqrt{(-4)^2 + (11)^2} \\ &= \sqrt{16 + 121} \\ &= \sqrt{137} = 11.7 \end{aligned}$$

$$\tan^{-1}\left(\frac{11}{-4}\right) = \cancel{70}^\circ \quad \cancel{109.9}^\circ \quad 70^\circ$$

$$z = z \Rightarrow z = -6$$

$$(11.7, \cancel{70}^\circ, -6)$$

cylindrical

$$\begin{aligned} r &= \sqrt{x^2 + y^2 + z^2} = \sqrt{(-4)^2 + (11.7)^2 + (-6)^2} \\ &= \sqrt{16 + 121 + 36} \end{aligned}$$

spherical

$$= 13.15$$

$$\begin{aligned} \theta &= \cos^{-1}\left(\frac{z}{r}\right) = \cos^{-1}\left(\frac{-6}{13.15}\right) \\ &= \cancel{22}^\circ \quad 117.16^\circ \end{aligned}$$

spherical

$$(\theta, \phi, \rho)$$

$$(13.15, \cancel{22}^\circ, \cancel{70}^\circ)$$

$$117.16$$

$$\cancel{109.9}^\circ \quad 70^\circ$$

QOC (ii)

cylindrical $\rightarrow$ Rectange

$$\rho = 5$$

$$\phi = 70^\circ$$

$$z = -4$$

$$x = \rho \cos \phi = 5 \cos 70^\circ = 5(0.34202) = 1.7101$$

$$y = \rho \sin \phi = 5 \sin 70^\circ = 5(0.939693) = 4.6985$$

$$z = -4$$

$$(x, y, z) = (1.7101, 4.6985, -4)$$

cylindrical $\rightarrow$ spherical

$$R = \sqrt{\rho^2 + z^2} \Rightarrow \sqrt{5^2 + (-4)^2} = \sqrt{25 + 16} = \sqrt{41} = 6.4031$$

$$\theta = \cos^{-1}\left(\frac{z}{R}\right) \Rightarrow \cos^{-1}\left(\frac{-4}{6.4031}\right) = \cos^{-1}(-0.624691) = 128.65^\circ$$

$$\phi = 70^\circ$$

$$(r, \phi, \theta) = (6.4031, 70^\circ, 128.65^\circ)$$

QOC (iii)

Rectangle

$$r = 5$$

$$\theta = 50^\circ$$

$$\phi = 100^\circ$$

$$x = r \sin \theta \cos \phi = 5 (\sin 50^\circ) (\cos 100^\circ)$$

$$= 5 (0.7660) (-0.1736) = -0.667$$

$$y = r \sin \theta \sin \phi = 5 (\sin 50^\circ) (\sin 100^\circ)$$

$$= 5 (0.7660) (0.9848) = 3.770$$

$$z = r \cos \theta = 5 (\cos 50^\circ) = 3.213$$

$$(x, y, z) = (-0.667, 3.770, 3.213) \rightarrow \text{degree}$$

$$= (3.30, -1.93, 3.21) \rightarrow \text{rad}$$

cylindrical

$$\begin{aligned} \rho &= r \sin \theta = 5 \sin(100^\circ) \\ &= 5(0.7660) \\ &= 3.8302 \end{aligned}$$

$$\phi = 100^\circ$$

~~0/pv~~

$$\begin{aligned} z &= r \cos \theta = 5 \cos 100^\circ \\ &= 3.21 \end{aligned}$$

~~$(\rho, \phi, z) =$~~

$$\begin{aligned} &(3.83, 100, 3.21) \rightarrow \text{degree} \\ &(-1.3, 100, 4.311) \rightarrow \text{rad} \end{aligned}$$